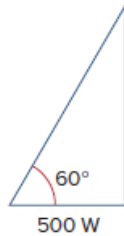


Problem

In the power triangle shown in Fig. 11.34(a), the reactive power is:

- (a) 1000 VAR leading
- (b) 1000 VAR lagging
- (c) 866 VAR leading
- (d) 866 VAR lagging

Figure 11.34(a):



Step-by-step solution

Step 1 of 2

Refer to Figure 11.34(a) in the textbook.

From the power triangle, the real power is,

$$P = 500 \text{ W}$$

Power factor angle is,

$$\theta = 60^\circ$$

Calculate the reactive power.

$$\begin{aligned} Q &= P \tan \theta \\ &= 500 \tan(60^\circ) \\ &= 866 \text{ VAR} \end{aligned}$$

Step 2 of 2

The reactive power is positive. Therefore, it is lagging.

Therefore, the reactive power, Q is 866 VAR lagging.

Therefore, the correct option is d.